

Semiconductor Defect Detection by Classical CV

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Reference-vs.-inspected comparison with rigid alignment, anomaly scoring, and peak-based defect extraction

Problem

Given a *reference* image and an *inspected* image of nominally identical chip regions, the goal is to output a **binary defect mask**. The main challenge is separating true defects from false positives caused by repeated geometry, strong edges, photometric differences, and small geometric misalignment.

Method

We use a modular classical computer-vision pipeline:

- **Preprocessing**: light denoising/stabilization.
- **Alignment**: constrained Euclidean registration (rotation + translation only), using coarse-to-fine angle search and translation estimation with structure-based scoring.
- **Photometric normalization**: align global intensity statistics.
- **Anomaly map**: continuous discrepancy map emphasizing structure and local contrast.
- **Peak-based extraction**: detect local maxima, score via anomaly strength and *peakness*, suppress edge responses, and apply NMS.
- **Binary mask refinement**: build final mask from connected support around accepted peaks with local refinement.

Why this works

- **Rigid alignment** removes dominant structured mismatch.
- **Structure-aware comparison** reduces sensitivity to illumination.
- **Peak selection** matches the localized nature of defects.
- **Edge suppression** eliminates dominant false positives.

Results

Figure 1 shows the final pipeline panels for the three cases.

The required assignment output — the binary defect mask — is explicitly shown in each panel as the second-to-last image labeled “Final Binary Defect Mask”.

- **Case 1**: three localized defects correctly detected.
- **Case 2**: defects preserved while suppressing structured clutter.
- **Case 3**: no detections, indicating correct rejection of false positives.

Assumptions

- Images differ only by small rigid misalignment.
- Defects are localized and produce strong local contrast.
- Background discrepancies are structured and extended.

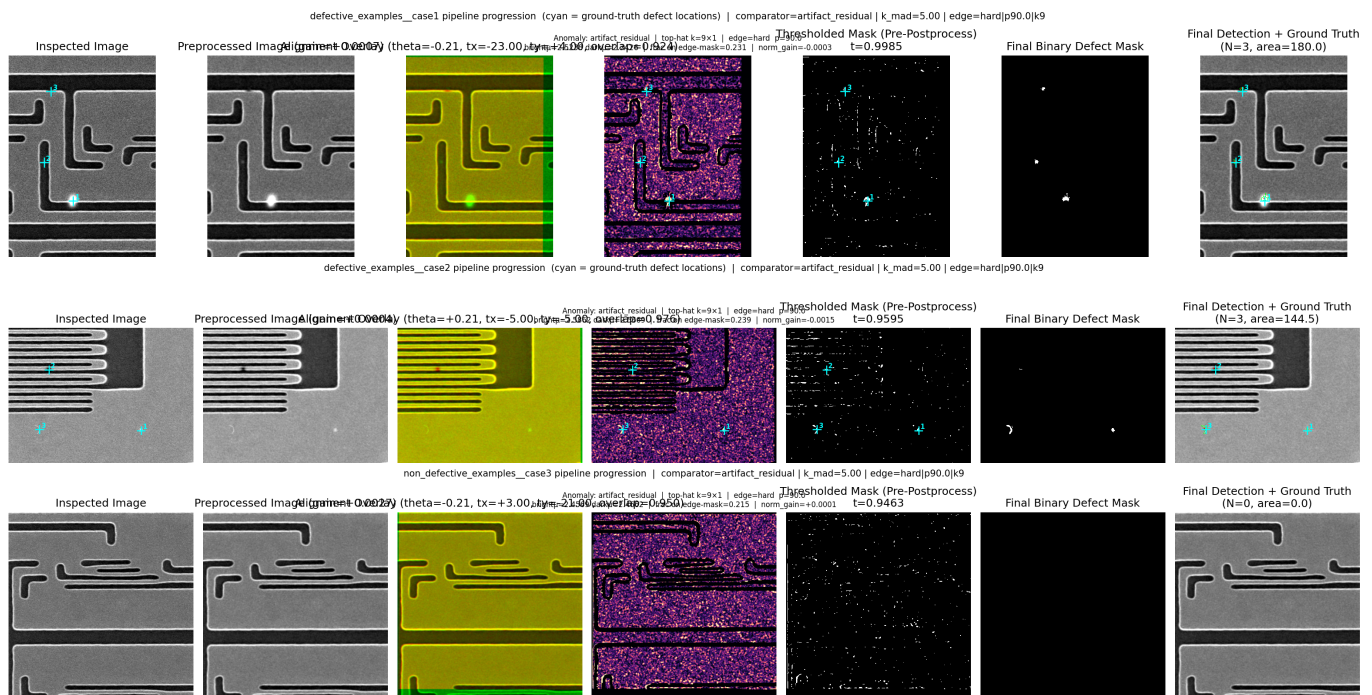


Figure 1: Final per-case pipeline visualizations. Each row shows: inspected image, preprocessing, alignment, anomaly map, thresholded mask, **final binary defect mask (second-to-last panel)**, and final detection with GT overlay.